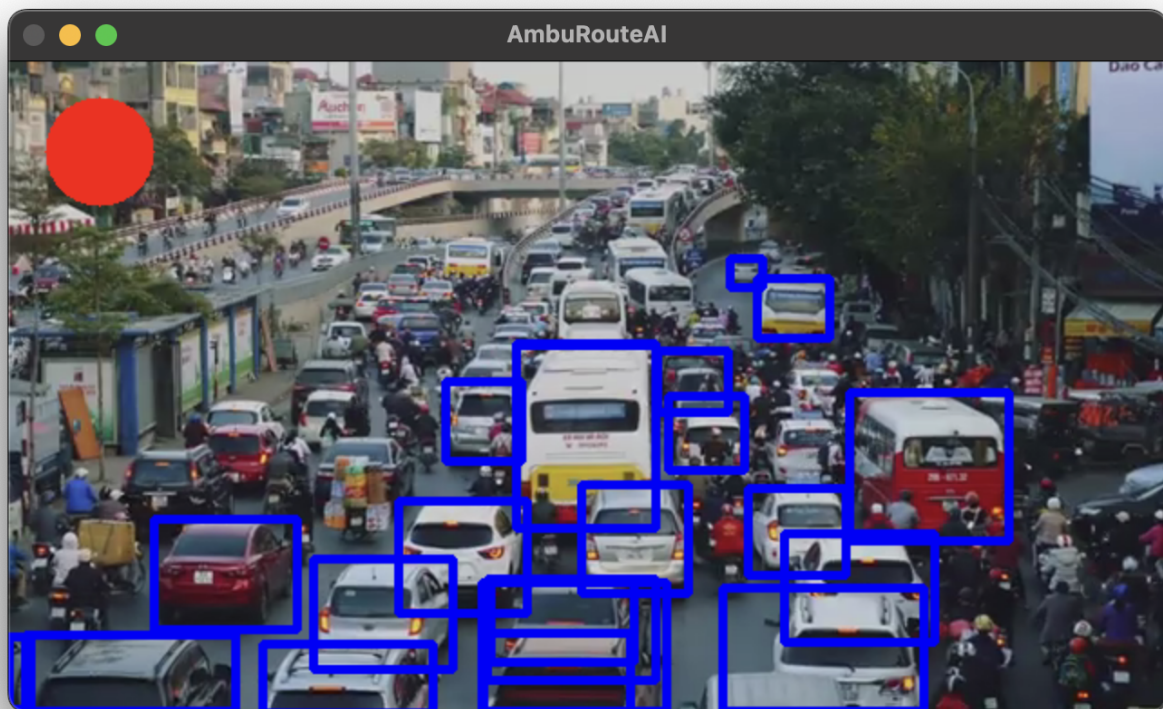


AmbuRouteAI



🚑 An AI-Powered Traffic Management for Emergency Vehicles.

🚦 AmbuRouteAI is an AI-powered traffic management system that detects ambulances in real-time and dynamically controls traffic signals to provide a clear path for emergency vehicles. Using YOLOv8 for object detection and OpenCV for traffic light simulation, this system helps reduce ambulance response time and enhances emergency healthcare logistics.

🔧 Features

✅ Real-time Ambulance Detection – Uses YOLOv8 AI to identify ambulances from live traffic feeds. ✅
Smart Traffic Signal Control – Dynamically changes traffic lights using OpenCV when an ambulance is detected. ✅
Live Video Processing – Works with webcams or traffic camera feeds to analyze road conditions. ✅
Seamless Integration – Can be extended to IoT-enabled smart city infrastructure. ✅
Scalable & Efficient – Future-ready for integration with Google Maps API for real-time route optimization.

🔧 Tech Stack

Component Technology/Tool
AI Model YOLOv8 (Ultralytics) Computer Vision OpenCV, NumPy Backend
Python Traffic Simulation OpenCV Live Video Input Webcam / CCTV Feed

How It Works

1. AI-Powered Detection – YOLOv8 identifies ambulances in real-time from traffic video feeds.
2. Traffic Light Control – If an ambulance is detected, the traffic light turns green; otherwise, it remains red.
3. Visual Alerts & UI – The system draws bounding boxes around detected ambulances and simulates traffic signals.
4. Extensibility – The project can integrate with Google Maps API & IoT sensors for smarter city-wide traffic control.  Installation & Setup

Run

Download [yolov8n.pt](#)

```
git clone https://github.com/mantreshkhurana/AmbuRouteAI.git
cd AmbuRouteAI
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python main.py
```

Future Enhancements

- ☐ Integrate with Google Maps API – Real-time traffic data for route optimization.
- ☐ IoT-Based Smart Traffic Signals – Connect with Raspberry Pi & Arduino for real-world applications.
- ☐ Hospital Alert System – Send emergency notifications to hospitals about incoming patients.
- ☐ Mobile App Interface – Develop an ambulance tracking app with live traffic updates.

Authors

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